

Systematization of the Nonequilibrium Integrated Phase (NEIP)

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We identify the Nonequilibrium Integrated Phase (NEIP), a nonequilibrium regime in which integratedness, multiscaleness, temporal recurrence, and sustained dissipation emerge in a coordinated manner not realized in any established framework. To characterize this phase, we introduce the Nonequilibrium Integrated Conditions (NEIC), a model-independent set of four macroscopic observables capturing these signatures. Using the Integrated Field Equation (IFE), a minimal stochastic field model combining gradient flow, solenoidal driving, nonlocal coupling, memory, and noise, we demonstrate that NEIC is satisfied within a finite and reproducible parameter region. A central result is that the NEIP boundary coincides with a singularity of the Fisher information metric constructed from $(A, d_s(k), G(T), \Sigma)$, providing a geometric criterion for nonequilibrium phase transitions. These results establish NEIP as a coherent, reproducible, and geometrically identifiable nonequilibrium phase occupying a previously unpopulated region of nonequilibrium phase space.

I. INTRODUCTION

Complex nonequilibrium systems, ranging from growing interfaces and driven lattice gases to active matter, often exhibit long-range correlation, nonlocal interaction, multiscale structure, and history dependence [1–4]. A wide range of theoretical frameworks—including Directed Percolation (DP) [5], Replica Symmetry Breaking (RSB) [6], Dynamical Mean Field Theory (DMFT) [7], the Martin – Siggia – Rose – Janssen – de Dominicis (MSRJD) formalism [8], and the Closed Time Path (CTP) formalism [9]—capture important aspects of nonequilibrium behavior. However, each framework typically realizes only a subset of the fundamental properties observed in complex nonequilibrium environments.

To highlight this conceptual gap, Table I summarizes the ability of these frameworks to realize five key properties: long-range correlation, multiscaleness, temporal memory, persistent nonequilibrium currents, and global integratedness. No existing framework simultaneously exhibits all five, despite their potential coexistence in physical systems such as dense bacterial suspensions or polar active fluids.

While many nonequilibrium systems exhibit one or more of these properties, no established framework is known to generate all of them simultaneously. Active matter models typically produce persistent probability currents but lack multiscaleness and non-Markovian recurrence [10]; driven lattice gases exhibit sustained nonequilibrium currents but possess single-scale spatial structure and memoryless dynamics; and nonlocal reaction – diffusion systems generate multiscaleness but do not exhibit coherent integratedness or persistent dissipation. In contrast, the Nonequilibrium Integrated Phase (NEIP) is defined by the simultaneous emergence of integratedness, multiscaleness, temporal recurrence, and sustained entropy production [11], captured respectively by

the observables $(A, d_s(k), G(T), \Sigma)$. This combination of macroscopic signatures does not appear in any existing nonequilibrium phase, positioning NEIP as a distinct and previously uncharacterized regime in nonequilibrium statistical mechanics.

Framework	LR corr.	Multi	Memory	Currents	Integr.
DP	✓	×	×	✓	×
RSB	✓	✓	×	×	×
DMFT	×	×	✓	×	×
MSRJD	✓	×	×	✓	×
CTP	✓	×	✓	×	×
NEIP	✓	✓	✓	✓	✓

TABLE I. Comparison of major nonequilibrium frameworks. NEIP uniquely satisfies all five fundamental properties simultaneously.

A central result of this work is that the NEIP boundary coincides with a singularity of the Fisher information metric [12–15] constructed from the macroscopic observable vector $(A, d_s(k), G(T), \Sigma)$. In our framework, the collapse of the metric determinant ($\det g \rightarrow 0$) corresponds to the divergence of the covariance matrix ($\det C \rightarrow \infty$), providing a geometric criterion for nonequilibrium phase transitions analogous to a diverging susceptibility. This establishes NEIP as a coherent, reproducible, and geometrically identifiable nonequilibrium phase.

From a geometric perspective, the NEIC observables serve as coarse-grained probes of the underlying information geometry of the nonequilibrium steady state. Integratedness and multiscaleness reflect global and scale-dependent features of the Fisher metric associated with the stationary distribution, while temporal recurrence in $G(T)$ captures non-Markovian structure in the corresponding statistical manifold. The entropy-production rate Σ quantifies the degree of metric asymmetry induced by probability currents, linking NEIP to geometric formulations of nonequilibrium thermodynamics.

Conceptually, NEIP provides a structural lens through

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which nonequilibrium organization can be understood independently of the microscopic mechanisms that generate it. Rather than relying on model-specific ingredients such as activity, external driving, or nonlocal interactions, NEIP characterizes nonequilibrium behavior in terms of macroscopic geometric and informational features that can be identified across diverse systems. This perspective highlights that integratedness, multiscale-ness, temporal recurrence, and sustained dissipation are not isolated phenomena but can emerge in a coordinated fashion to form a distinct organizational regime.

The remainder of this paper formulates the Integrated Field Equation (IFE) as a minimal constructive example, defines the Nonequilibrium Integrated Conditions (NEIC), presents numerical evidence for the existence of NEIP, and demonstrates that the Fisher metric singularity provides a precise and reproducible criterion for identifying the NEIP boundary.

II. IFE MINIMAL MODEL

We consider a scalar field $\phi(x, t)$ on a one-dimensional domain with periodic boundary conditions. While strict one-dimensional equilibrium systems with short-range interactions cannot exhibit spontaneous symmetry breaking at finite noise levels due to the Mermin – Wagner theorem, the inclusion of nonlocal coupling and nonequilibrium driving in our framework bypasses this restriction, allowing for the emergence of macroscopic order.

The Integrated Field Equation (IFE) is defined as

$$\begin{aligned} \frac{d\phi(x, t)}{dt} = & -\frac{\delta F[\phi]}{\delta \phi(x, t)} + S[\phi](x, t) \\ & + \int dy J(x, y)\phi(y, t) \\ & + \int_0^t ds K(t-s)\phi(x, s) \\ & + \xi(x, t), \end{aligned} \quad (1)$$

combining gradient flow, solenoidal driving, nonlocal coupling, temporal memory, and stochastic forcing. These components collectively generate the macroscopic signatures captured by NEIC.

A. Gradient flow

The gradient-flow component is derived from the free-energy functional

$$F[\phi] = \int dx \left[\frac{a}{2}(\partial_x \phi)^2 + \frac{b}{4}(\phi^2 - 1)^2 \right]. \quad (2)$$

The double-well potential produces multistability, enabling exploration of multiple metastable configurations rather than relaxation to a trivial single minimum.

B. Solenoidal driving

The nonequilibrium driving term

$$S[\phi](x, t) = \alpha \partial_x \phi(x, t) \quad (3)$$

breaks detailed balance and generates rotational probability currents, maintaining the system in a genuine nonequilibrium steady state.

C. Nonlocal coupling

Spatial nonlocality is introduced through the Gaussian kernel

$$J(x, y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-y)^2}{2\sigma^2}\right]. \quad (4)$$

This term produces long-range spatial interactions and generates scale-dependent structure in the power spectrum.

D. Memory kernel

Temporal nonlocality is introduced via an exponential memory kernel

$$K(t) = Be^{-\gamma t}. \quad (5)$$

This produces history-dependent dynamics and long-time correlation, including oscillatory recurrence.

E. Noise

The stochastic forcing

$$\langle \xi(x, t)\xi(x', t') \rangle = \sigma^2 \delta(x - x')\delta(t - t') \quad (6)$$

enables exploration of metastable states and prevents trapping in a single configuration.

F. Minimality of the IFE

Each component of the IFE contributes a distinct aspect of the macroscopic behavior associated with NEIP. Removing any one of them eliminates at least one NEIC signature, establishing the IFE as a minimal constructive model capable of realizing the Nonequilibrium Integrated Phase.

III. MATHEMATICAL DEFINITION OF NEIC

The four NEIC observables constitute a minimal and nonredundant set: each captures a distinct macroscopic

signature of NEIP, and removing any one of them eliminates a necessary aspect of the coordinated nonequilibrium behavior. Long-range spatial correlation and global integratedness are jointly represented by the integrated order parameter A , while multiscaleness, temporal memory, and persistent probability currents map directly onto $d_s(k)$, $G(T)$, and Σ .

No established nonequilibrium framework satisfies all four conditions simultaneously: active matter lacks multiscaleness and recurrence, driven lattice gases lack memory, and nonlocal reaction – diffusion systems lack coherent integratedness and sustained dissipation. NEIP is therefore not a reinterpretation of any known phase but a distinct regime defined by the simultaneous emergence of these macroscopic features.

We define the four-dimensional observable vector

$$X = (A, d_s(k), G(T), \Sigma), \quad (7)$$

which provides a complete and mathematically independent basis for characterizing NEIP. Because the NEIC conditions are formulated entirely in terms of macroscopic observables, they apply independently of the microscopic details of the underlying dynamics, making NEIP a genuinely model-agnostic nonequilibrium phase.

NEIC is satisfied only when all four observables exhibit their characteristic signatures:

$$\text{NEIC} = \left\{ \langle |A| \rangle > 0, \quad \text{Var}[d_s(k)] > 0, \right. \\ \left. G(T) \text{ shows recurrence}, \quad \Sigma > 0 \right\}. \quad (8)$$

Although the four conditions are independent, their breakdown typically follows a hierarchical pattern: sustained dissipation Σ is the most robust, temporal recurrence $G(T)$ weakens first, and multiscaleness $d_s(k)$ collapses before global integratedness A . This hierarchy clarifies how NEIP transitions into less organized nonequilibrium regimes as control parameters are varied. The four NEIC conditions are not only necessary but jointly sufficient for identifying NEIP. Each observable constrains a distinct axis of the macroscopic state space: A fixes the global mode, $d_s(k)$ determines the scale hierarchy, $G(T)$ specifies temporal organization, and Σ encodes the degree of nonequilibrium driving. When all four conditions are satisfied, the observable vector $X = (A, d_s(k), G(T), \Sigma)$ occupies a compact region of state space in which no further independent macroscopic degree of freedom remains unconstrained. Conversely, violating any single condition enlarges the accessible manifold, allowing trajectories that leave the coordinated regime characteristic of NEIP. In this sense, NEIC forms a minimal and complete set: it excludes all non-integrated nonequilibrium behavior while admitting exactly the configurations that constitute the Nonequilibrium Integrated Phase.

IV. NUMERICAL RESULTS

We numerically integrate the IFE using periodic boundary conditions, a time step $\Delta t = 0.01$, and a spatial resolution $\Delta x = 0.02$. Noise realizations, initial conditions, and sampling procedures are fixed across all experiments to ensure statistical reproducibility.

A. Numerical rigor and finite-size scaling

The stochastic differential equation (Eq. 1) is integrated using the Euler – Maruyama scheme, with the memory kernel truncated at $T_{\text{cutoff}} = 10/\gamma$ to retain non-Markovian effects while maintaining computational feasibility. The dissipation rate Σ is evaluated using the Ito convention to avoid spurious noise-induced drifts.

The spectral dimension $d_s(k)$ is computed using logarithmic smoothing, with error bands obtained from 10^3 independent realizations. To verify that the emergence of $\langle |A| \rangle > 0$ and the Fisher metric singularity are not finite-size artifacts, we perform finite-size scaling across system sizes $N = 512 - 4096$. The extracted thresholds obey $\alpha_c(N) = \alpha_c(\infty) + cN^{-1/\nu}$, confirming that the NEIP boundary remains sharp in the thermodynamic limit.

Simulations initialized from distinct states (white noise, localized sine wave, uniform polarized state) converge to identical steady-state statistics, demonstrating complete insensitivity to initial conditions.

B. Integratedness: Emergence of macroscopic order

Figure 1 shows the time evolution of $\phi(x, t)$ in the NEIP region. The system exhibits intermittent exploration and repeated revisits to metastable configurations, preventing relaxation to the trivial minimum $\phi = 0$. As a result, the time-averaged magnitude $\langle |A| \rangle$ remains strictly positive. The corresponding time-averaged field profile (Fig. ??) displays broad, coherent spatial structure, confirming robust integratedness.

C. Multiscaleness: Scale-dependent spectral dimension

The spatial power spectrum $S(k)$ (Fig. 6) exhibits a broad distribution with a characteristic wavelength $\lambda_{\text{space}} \sim 0.08 - 0.3$. The spectral dimension $d_s(k)$ varies significantly with scale, distinguishing NEIP from single-scale roughening processes such as Edwards – Wilkinson or KPZ dynamics and indicating intrinsic multiscale spatial organization.

D. Long-time correlation: Temporal memory

The two-point Green function $G(T)$ (Fig. 5) shows slow envelope decay with superimposed oscillatory recurrence. The frequency spectrum exhibits a dominant peak at $f_{\text{peak}} = 0.2$, corresponding to a recurrence period $T_{\text{peak}} = 5.0$. The envelope decays exponentially with rate $\gamma = 0.971$, yielding a memory time $1/\gamma = 1.03$. The co-existence of short-term decay and mid-scale recurrence confirms robust non-Markovian temporal memory.

E. Nonequilibrium nature: Sustained dissipation

The macroscopic dissipation rate Σ , computed from the solenoidal probability current, remains strictly positive throughout the NEIP region. This indicates continuous entropy production and persistent violation of detailed balance, confirming that NEIP corresponds to a genuine nonequilibrium steady state.

F. Construction of the NEIP phase diagram

By scanning the parameter space of the IFE (e.g., varying driving strength α and noise intensity σ), we identify regions where all four NEIC criteria are simultaneously satisfied. The resulting phase diagram divides into three regimes: a strongly fluctuating exploration phase, a frozen trapped phase, and the NEIP region. NEIP is uniquely characterized by $\langle |A| \rangle > 0$, scale-dependent $d_s(k)$, temporal recurrence, and sustained dissipation.

V. FISHER METRIC AND THE NEIP BOUNDARY

The Fisher information metric provides a geometric measure of distinguishability between probability distributions in observable space. In the context of nonequilibrium phase transitions, metric singularities indicate that fluctuations in one observable become fully constrained by the others, revealing a collapse of the effective dimensionality of the observable manifold. We show that the boundary of the Nonequilibrium Integrated Phase (NEIP) is precisely characterized by such a geometric singularity.

A. Construction and Standardization of the Fisher Metric

Let $X = (A, d_s(k), G(T), \Sigma)$ denote the four-dimensional observable vector defined by the Nonequilibrium Integrated Conditions (NEIC). Because these observables possess fundamentally different physical units and scales, the determinant of the raw covariance matrix is not invariant under scaling transformations. To ensure

coordinate-scale invariance, we standardize the observables using their z-scores:

$$\tilde{X}_i = \frac{X_i - \langle X_i \rangle}{\sigma_i}, \quad (9)$$

where σ_i is the standard deviation of X_i .

Assuming that the macroscopic observables follow an approximately Gaussian distribution in the steady state due to the central limit theorem for large systems, the Fisher information metric can be well approximated by the inverse of the correlation matrix R_{ij} :

$$g_{ij} \approx (R^{-1})_{ij}, \quad R_{ij} = \langle \tilde{X}_i \tilde{X}_j \rangle. \quad (10)$$

While strong non-Gaussian fluctuations may appear strictly at the critical point, this inverse-covariance approximation robustly captures the leading-order divergence of macroscopic susceptibilities near the phase boundary and provides a reliable geometric indicator for the transition.

B. Numerical Regularization, Stability, and ϵ -Sweep Analysis

As the system approaches the NEIP boundary, macroscopic fluctuations diverge and the correlation matrix R becomes nearly singular. In this regime, direct computation of R^{-1} becomes numerically ill-posed, leading to an explosion of the condition number $\kappa(R)$.

To maintain numerical stability and to demonstrate that the observed singularity is a genuine physical phenomenon rather than a numerical artifact, we apply Tikhonov regularization (ridge regression) by introducing a regularization parameter ϵ :

$$g_{ij}(\epsilon) = [(R + \epsilon I)^{-1}]_{ij}, \quad (11)$$

where I is the identity matrix. To verify robustness against the choice of ϵ , we performed a systematic ϵ -sweep across four orders of magnitude, $\epsilon \in [10^{-8}, 10^{-4}]$.

Across all values of ϵ , the estimated critical threshold α_c^{Fisher} —defined by the collapse of the metric determinant $\det g(\epsilon) \rightarrow 0$, the vanishing of the minimum eigenvalue $\lambda_{\min}(g)$, and the sharp peak in the condition number $\kappa(R)$ —remains remarkably stable. Bootstrap resampling (10^3 iterations) confirms that the uncertainty in α_c^{Fisher} is below 0.5%, demonstrating that the Fisher-metric singularity is an invariant geometric property of the system.

C. Geometric Signature of the NEIP Boundary

Within the deep NEIP region, the regularized Fisher metric g_{ij} is positive definite and all eigenvalues remain finite. However, as the system approaches the NEIP

boundary, macroscopic fluctuations diverge and the observables develop increasingly strong correlations. Because the Fisher metric satisfies $g \approx R^{-1}$, the singularity of the correlation matrix,

$$\det R \rightarrow 0, \quad (12)$$

directly produces a geometric collapse of the Fisher metric determinant:

$$\det g \rightarrow 0. \quad (13)$$

Physically, this geometric singularity implies that fluctuations in one property (e.g., integratedness) become fully determined by combinations of the others (e.g., multiscaleness and dissipation). NEIP thus behaves as an effectively low-dimensional manifold embedded in the higher-dimensional observable space.

D. Quantitative Agreement with the Phenomenological Boundary

A crucial test for this geometric approach is its consistency with the phenomenological definition of NEIP. We independently determined the transition point α_c^{NEIC} where the simultaneous satisfaction of the four NEIC conditions completely breaks down. Separately, we tracked the Fisher metric determinant to identify α_c^{Fisher} where $\det g$ collapses.

Our numerical analysis confirms that α_c^{NEIC} and α_c^{Fisher} coincide quantitatively within a narrow error band ($\Delta\alpha/\alpha < 2\%$). This agreement demonstrates that the Fisher-metric singularity is not a numerical artifact but a robust, model-independent criterion for identifying nonequilibrium phase transitions.

As shown in Fig. 8, while the absolute values near the transition point show a minor, controlled dependence on ϵ , the estimated critical threshold α_c^{Fisher} (defined by $\det g(\epsilon) \rightarrow 0$, the vanishing of the minimum eigenvalue $\lambda_{\min} \rightarrow 0$, and the sharp peak in the condition number $\kappa(R)$) remains remarkably stable. Furthermore, bootstrap resampling (10^3 iterations) confirms that the uncertainty of α_c^{Fisher} due to finite-sample fluctuations and regularization bias yields a 95% confidence interval of less than 0.5%. This guarantees that the Fisher metric collapse is an invariant geometric property of the system.

E. Geometric Signature of the NEIP Boundary

Within the deep NEIP region, the regularized Fisher metric g_{ij} is positive definite, and its eigenvalues remain finite and strictly positive. However, as the system approaches the boundary of the NEIP, macroscopic fluctuations diverge and the observables develop increasingly strong correlations. Because the Fisher metric is approximated by the inverse of the correlation matrix ($g \approx R^{-1}$),

the divergence of the largest eigenvalue of R (corresponding to $\det R \rightarrow \infty$) directly drives the collapse of the Fisher metric determinant:

$$\det g \rightarrow 0. \quad (14)$$

Physically, this geometric singularity implies that fluctuations in one property (e.g., integratedness) become fully determined by combinations of the others (e.g., multiscaleness and dissipation). NEIP thus behaves as a low-dimensional manifold embedded in the higher-dimensional observable space.

F. Quantitative Agreement with the Phenomenological Boundary

A crucial test for this geometric approach is its consistency with the phenomenological definition of NEIP. We independently determined the transition point α_c^{NEIC} where the simultaneous satisfaction of the four NEIC conditions completely breaks down. Separately, we tracked the metric determinant to identify α_c^{Fisher} where $\det g$ falls below a defined numerical threshold bounded by the regularization error.

Our numerical analysis confirms that α_c^{NEIC} and α_c^{Fisher} coincide quantitatively within a narrow error band ($\Delta\alpha/\alpha < 2\%$). This precise agreement confirms that the Fisher metric singularity is not a numerical artifact but a robust, model-independent criterion for identifying the nonequilibrium phase transition.

VI. CONCLUSION

We have identified the Nonequilibrium Integrated Phase (NEIP), a distinct nonequilibrium regime characterized by the coordinated emergence of integratedness, multiscaleness, temporal recurrence, and sustained dissipation. By formulating the Nonequilibrium Integrated Conditions (NEIC), we provided a model-independent set of macroscopic criteria that precisely delineate this phase. The Integrated Field Equation (IFE) serves as a minimal constructive example demonstrating that NEIP can be realized within a finite and reproducible parameter region.

A central result of this work is that the NEIP boundary coincides with a singularity of the Fisher information metric constructed from $(A, d_s(k), G(T), \Sigma)$. The collapse of the metric determinant provides a geometric criterion for nonequilibrium phase transitions, linking NEIP to a diverging susceptibility in information geometry and revealing that NEIP occupies a previously unpopulated region in nonequilibrium phase space.

Our numerical results confirm that NEIP is robust under changes in system size, initial conditions, and noise realizations, and that its critical behavior is reproducible across a broad parameter range. These findings establish NEIP as a coherent and geometrically identifiable

nonequilibrium phase, offering a structural framework for understanding how complex driven systems can exhibit coordinated large-scale organization while remaining fundamentally dissipative.

The NEIP framework provides a foundation for future investigations of nonequilibrium phases in higher dimensions, systems with additional symmetries, and models with more complex interaction kernels. More broadly, it suggests that coordinated spatial, temporal, and dynamical structure may arise generically in driven stochastic systems, opening a path toward a unified geometric classification of nonequilibrium organization.

Appendix A: Numerical Methods and Quantitative Criteria

This appendix details the numerical procedures, parameter ranges, and quantitative thresholds used to evaluate the Nonequilibrium Integrated Conditions (NEIC) and construct the phase diagram, ensuring full computational reproducibility.

1. Parameter Spaces and Physical Relevance

The Integrated Field Equation (Eq. 1) was systematically integrated over the following parameter ranges:

$$\alpha \in [0, 1.5], \quad \sigma \in [0.05, 0.5], \quad \gamma \in [0.1, 2.0], \quad \sigma_\xi \in [0.01, 0.5] \quad (\text{A1})$$

These ranges were chosen to reflect physically realizable regimes: α controls the Péclet number equivalent for active driving, σ scales the nonlocal interaction range relative to the system size ($L = 40.96$), and γ dictates the viscoelastic relaxation rate. The noise intensity σ_ξ was calibrated to induce barrier crossing in the double-well potential without overwhelming the deterministic dynamics.

2. Quantitative Thresholds for NEIC

To prevent subjective classification and reliably construct the phase diagram, the four NEIC observables were subjected to strict numerical thresholds averaged over $T_{\text{sample}} = 8 \times 10^4$:

1. **Integratedness:** $\langle |A| \rangle > 0.05$. This threshold safely excludes $O(1/\sqrt{N})$ thermal fluctuations around $A = 0$.
2. **Multiscaleness:** The variance of the local slope of the spectral dimension must satisfy $\text{Var}[d_s(k)] > 0.1$ over the inertial range $k \in [2\pi/L, 20\pi/L]$, ensuring genuine scale dependence rather than simple power-law scaling.

3. **Long-time correlation (Memory):** The envelope of $G(T)$ is fitted to $G_{\text{env}}(T) \sim e^{-\gamma_{\text{eff}} T}$. We require the secondary oscillatory peak height to exceed $0.1 \times G(0)$, and the memory time $\tau_m = 1/\gamma_{\text{eff}}$ to satisfy $\tau_m > 1.0$.

4. **Dissipation:** $\Sigma > 10^{-3}$, well above the numerical integration noise floor, ensuring persistent entropy production.

From the standpoint of information geometry, the NEIC observables can be related to geometric features of the Fokker – Planck operator governing the nonequilibrium steady state. The integrated order parameter A reflects coherent structure in the stationary probability density and corresponds to global organization in the associated Fisher metric. The multiscale spectrum $d_s(k)$ probes scale-dependent curvature of this statistical manifold, while the recurrence function $G(T)$ captures non-Markovian structure encoded in the temporal evolution of probability flows. The entropy-production rate Σ quantifies the antisymmetric component of the probability current, linking NEIP to geometric formulations of irreversibility based on information flow. These connections show that NEIP occupies a distinct region in the information-geometric landscape of driven stochastic systems, providing a mathematical foundation for the structural criteria used in NEIC.

A useful geometric decomposition of the probability flow associated with the nonequilibrium steady state is obtained by splitting the current $J(x)$ into symmetric and antisymmetric components with respect to the Fokker – Planck operator. The symmetric part corresponds to gradient flows that preserve detailed balance and determines the reversible deformation of the stationary distribution, while the antisymmetric part encodes circulating probability fluxes that break time-reversal symmetry and generate positive entropy production. In this representation, the NEIC observable Σ measures precisely the magnitude of the antisymmetric component, linking NEIP to geometric formulations of irreversibility based on information flow. This decomposition clarifies how integrated structure, multiscaleness, and temporal recurrence can coexist with persistent circulation in probability space, providing a mathematical basis for the coordinated nonequilibrium behavior characteristic of NEIP.

3. Phase Diagram Construction Algorithm

The phase diagram (Fig. ??) was constructed using a grid-search algorithm across the (α, σ_ξ) plane. For each grid point, 10^2 independent realizations were simulated. The system was classified into the NEIP only if the ensemble average of the observables simultaneously passed all four thresholds defined above. The boundary was independently verified by tracing the roots of

$\det g(\epsilon) - \delta_c = 0$, where $\delta_c = 10^{-4}$ is the regularization-induced offset.

4. Quantitative Proof of Minimality and Ablation Phase Diagrams

To quantitatively establish that the IFE is the minimal model for NEIP, we performed systematic ablation studies by selectively neutralizing individual mathematical components. The model was re-evaluated across a coarse two-dimensional parameter grid in the (α, σ_ξ) plane under three modified conditions:

1. **Without Solenoidal Driving** ($\alpha = 0$): The dissipation rate strictly collapses to $\Sigma = O(10^{-8})$, well below the threshold, eliminating the nonequilibrium steady current across the entire parameter space.
2. **Without Nonlocal Coupling** ($J(x, y) \rightarrow \delta(x - y)$): The spatial interaction becomes purely local, resulting in $\text{Var}[d_s(k)] < 10^{-3}$, which completely destroys the multiscale spatial structure.
3. **Without Memory Kernel** ($K(t) \rightarrow \delta(t)$): The temporal nonlocality is removed, universally reducing the effective memory time to $\tau_m < 0.05$ and suppressing all oscillatory recurrence in $G(T)$.

As visualized by the ablation phase diagrams in Fig. ??, removing any single core mechanism causes the entire NEIP parameter volume to vanish. These control experiments rigorously prove that every component of the IFE is strictly necessary and jointly sufficient, confirming the true minimality of the model.

5. Estimation Procedures for the Spectral Dimension $d_s(k)$

To evaluate the scale-dependent spectral dimension $d_s(k) = -d \log P(k) / d \log k$, the spatial power spectrum $P(k)$ was computed via discrete Fourier transforms averaged over 10^3 steady-state configurations. To mitigate high-frequency numerical noise, a moving logarithmic sliding window of width $\Delta(\log k) = 0.15$ was applied. The local slope $d_s(k)$ was extracted via linear regression within each window over the inertial range, and statistical error bands were rigorously propagated to confirm that the observed scale-dependence exceeds the regression uncertainty.

Sensitivity Analysis of NEIC Thresholds (Fig. S1)

The Nonequilibrium Integrated Conditions (NEIC) consist of four observables:

$$X = (A, d_s(k), G(T), \Sigma),$$

each of which must exhibit its characteristic nonequilibrium signature for NEIP to be satisfied. Although the main text uses natural thresholds (e.g., $\langle |A| \rangle > 0$, $\text{Var}[d_s(k)] > 0$, recurrence in $G(T)$, and $\Sigma > 0$), it is essential to demonstrate that the NEIP boundary is not an artifact of specific numerical thresholds.

To test robustness, we systematically varied the NEIC thresholds by multiplicative factors:

$$\theta \in \{0.5, 0.75, 1.0, 1.25, 1.5\},$$

corresponding to $\pm 50\%$ variations around the default criteria. For each θ , we recomputed the NEIP region and extracted the critical driving strength $\alpha_c^{\text{NEIC}}(\theta)$.

The thresholds were modified as follows:

- Integratedness: $\langle |A| \rangle > \theta A_{\text{base}}$,
- Multiscaleness: $\text{Var}[d_s(k)] > \theta V_{\text{base}}$,
- Temporal memory: recurrence amplitude $R_{\text{peak}} > \theta R_{\text{base}}$,
- Dissipation: $\Sigma > \theta \Sigma_{\text{base}}$.

Figure S1 shows the resulting dependence of the NEIP boundary on θ .

Across all threshold variations, the critical point satisfies:

$$\frac{\Delta \alpha_c}{\alpha_c} < 2\%,$$

and the NEIP phase volume changes by less than 5%. This demonstrates that the NEIP boundary is not sensitive to the precise numerical values used in the NEIC criteria.

The stability of $\alpha_c^{\text{NEIC}}(\theta)$ under threshold variation confirms that NEIP is a genuine nonequilibrium phase rather than a threshold-induced artifact. Combined with the Fisher-metric analysis (Appendix A4 – A6), this establishes the robustness of the NEIP boundary from both phenomenological and geometric perspectives.

Finite-Size Scaling and Thermodynamic Extrapolation (Fig. S3)

To verify that the NEIP boundary is not a finite-size artifact, we performed systematic simulations across four system sizes:

$$N \in \{512, 1024, 2048, 4096\}.$$

For each N , we extracted the critical driving strength $\alpha_c(N)$ at which the NEIC conditions cease to be simultaneously satisfied. The resulting finite-size dependence is shown in Fig. S3(a).

The finite-size dependence of the critical point is well described by the scaling relation:

$$\alpha_c(N) = \alpha_c(\infty) + cN^{-1/\nu}, \quad (\text{A2})$$

where c is a nonuniversal constant and ν is the effective correlation length exponent. Fitting Eq. (A2) to the numerical data yields:

$$\alpha_c(\infty) = \alpha_c(N = 4096) \pm 0.8\%.$$

The extrapolated thermodynamic-limit value $\alpha_c(\infty)$ agrees with the Fisher-metric critical point α_c^{Fisher} within:

$$\Delta\alpha/\alpha < 2\%.$$

Furthermore, Fig. S3(c) shows that the NEIP phase volume in (α, σ_ξ) space remains finite and stable across all system sizes, demonstrating that NEIP persists in the thermodynamic limit.

These results confirm that the NEIP boundary is not a finite-size artifact and that the phase remains well-defined as $N \rightarrow \infty$. Combined with the Fisher-metric analysis (Appendix A4 – A6) and the NEIC threshold sensitivity analysis (Appendix A7), this establishes the robustness of the NEIP phase from phenomenological, geometric, and thermodynamic perspectives. In finite systems, the approach to the NEIP boundary is accompanied by characteristic finite-size scaling behavior. The recurrence amplitude $G(T)$ exhibits a crossover from power-law decay to exponential suppression at a temporal scale $T_{\text{fs}} \sim L^z$, where L is the system size and z an effective dynamical exponent. Likewise, the multiscale spectrum $d_s(k)$ shows a truncation of its power-law range at a wavenumber $k_{\text{fs}} \sim L^{-1}$, reflecting the loss of long-range scale coupling in finite domains. These finite-size effects smooth the transition but preserve the underlying scaling structure, allowing proximity to the NEIP boundary to be quantified through the collapse of $G(T)$ and $d_s(k)$ when rescaled by T_{fs} and k_{fs} , respectively. Close to the NEIP boundary, the NEIC observables exhibit scaling behavior that can be characterized through effective critical exponents. The recurrence amplitude $G(T)$ decays as $G(T) \sim T^{-\gamma}$ over an intermediate temporal range, with γ increasing as recurrence weakens. Similarly, the multiscale spectrum approaches a power-law form $d_s(k) \sim k^{-\beta}$, where the exponent β decreases as scale coupling collapses. These exponents do not represent universal critical indices in the equilibrium sense, but they quantify how recurrence and multiscale-ness are suppressed near the transition and thus provide a geometric measure of proximity to the NEIP boundary. A geometric interpretation of the Fisher-metric singularity can be obtained by noting that, for standardized observables $\tilde{X}_i$, the Fisher metric reduces to the inverse of the correlation matrix, $g_{ij} = (R^{-1})_{ij}$. Let $\lambda_{\text{max}}(R)$ denote the largest eigenvalue of R . Near the NEIP boundary, the macroscopic observables become increasingly constrained by one another, causing $\lambda_{\text{max}}(R)$ to diverge and the remaining eigenvalues to collapse. Because $\det g = 1/\det R$, the divergence of $\lambda_{\text{max}}(R)$ produces a geometric degeneracy,

$$\det g \sim \lambda_{\text{max}}(R)^{-1} \rightarrow 0,$$

indicating that fluctuations in one observable become fully determined by a lower-dimensional subspace of the others. Equivalently, the metric volume element shrinks as

$$\text{Vol}(g) = \sqrt{\det g} \rightarrow 0,$$

revealing that the observable manifold loses rank and collapses onto a lower-dimensional geometric structure. This provides a mathematically precise criterion for nonequilibrium phase transitions: NEIP is the unique region in parameter space where the Fisher metric remains nondegenerate, and the NEIP boundary is exactly the locus where the metric volume vanishes. A complementary geometric signature of the NEIP boundary is provided by the Ricci curvature of the Fisher metric. For the observable manifold with metric $g_{ij} = (R^{-1})_{ij}$, the Ricci scalar R is constructed from the Levi-Civita connection Γ_{ij}^k and the Riemann tensor $R^k_{ij\ell}$. Near the NEIP boundary, the largest eigenvalue of the correlation matrix diverges, causing g_{ij} to lose rank. Because the Ricci scalar contains contractions of g^{ij} with derivatives of g_{ij} , this degeneracy produces a curvature blow-up,

$$R \sim g^{ij} \partial_i \partial_j \log \det g \rightarrow +\infty,$$

indicating that the statistical manifold becomes increasingly “pinched” along the collapsing directions of the metric. This curvature divergence provides an invariant geometric marker of the NEIP boundary: the phase corresponds to the region where the Fisher metric is smooth and nondegenerate, while the transition occurs precisely where the Ricci curvature becomes singular.

6. Higher-Dimensional and Quantum Generalizations

In higher spatial dimensions and in quantum open systems the NEIP phenomenology extends under natural generalizations of the NEIC observables. In $d > 1$ the spectral dimension becomes anisotropic, $d_s(\mathbf{k})$, and multiscale-ness is diagnosed by direction-dependent scaling of $d_s(\mathbf{k})$ together with anisotropic structure factors; nonlocal coupling $J(\mathbf{x}, \mathbf{y})$ and solenoidal driving generalize to tensorial kernels and divergence-free vector fields whose spatial range and amplitude control interscale coupling. Finite-size scaling acquires dimension-dependent exponents (e.g. $T_{\text{fs}} \sim L^{z(d)}$), so measured crossover laws for $G(T)$ and d_s are generically nonuniversal and depend on d and the decay of J . In the quantum setting the classical Fisher metric is replaced by the quantum Fisher information metric on the manifold of steady-state density operators and the entropy-production rate is obtained from the dissipator; metric degeneracy ($\det g \rightarrow 0$) and curvature blow-up remain invariant indicators of a collapse of independent macroscopic degrees of freedom, thus defining a quantum NEIP boundary. Numerical and perturbative evidence shows that, provided long-range coupling and

non-Markovian memory are sufficiently strong relative to local relaxation, the NEIP region persists in 2D and 3D, although the precise location and scaling exponents of the boundary shift with dimensionality and microscopic dispersion relations.

Ablation Analysis of the IFE Components (Fig. 7)

To demonstrate that each component of the Integrated Field Equation (IFE) is necessary for generating the Nonequilibrium Integrated Phase (NEIP), we performed an ablation analysis in which individual mechanisms were removed one at a time. For each ablated model, we scanned the parameter space (α, σ_ξ) on a coarse grid (20×20) and evaluated whether the Nonequilibrium Integrated Conditions (NEIC) were simultaneously satisfied.

We considered the following ablated variants:

- (i) **No solenoidal driving:** $S[\phi] = 0$
- (ii) **No nonlocal coupling:** $J(x, y) \rightarrow \delta(x - y)$
- (iii) **No memory kernel:** $K(t) = 0$
- (iv) **No noise:** $\xi(x, t) = 0$

For comparison, the full IFE model (all mechanisms present) was evaluated on the same parameter grid.

Figure 7 shows the resulting phase diagrams for the full model and each ablated variant.

Across all ablated variants, the NEIP region disappears completely:

$$\text{NEIP}_{\text{ablated}} = \emptyset.$$

This confirms the following:

- Removing solenoidal driving eliminates nonequilibrium currents ($\Sigma = 0$).
- Removing nonlocal coupling eliminates multiscale-ness ($\text{Var}[d_s(k)] \rightarrow 0$).
- Removing the memory kernel eliminates temporal recurrence in $G(T)$.
- Removing noise eliminates exploration, suppressing integratedness ($\langle |A| \rangle \rightarrow 0$).

Thus, each mechanism in the IFE is necessary for satisfying at least one NEIC condition, and removing any single component collapses the NEIP entirely. Combined with the Fisher-metric analysis (Appendix A4 – A6), the NEIC threshold sensitivity analysis (Appendix A7), and the finite-size scaling results (Appendix A8), this ablation study provides definitive evidence that the IFE is a minimal and physically interpretable model for realizing the Nonequilibrium Integrated Phase.

Distinction from Existing Nonequilibrium Systems

Although many nonequilibrium systems exhibit one or more of the phenomenological properties associated with NEIP, none of the widely studied frameworks satisfy all four NEIC conditions simultaneously. Here we clarify, using the NEIC observables, why NEIP is not a relabeling of known nonequilibrium phases such as active matter, driven lattice gases, or nonlocal reaction – diffusion systems.

Active matter (e.g., polar flocks, active colloids). Active systems typically exhibit persistent probability currents ($\Sigma > 0$) and long-range correlation in certain regimes. However, their spatial structure is dominated by a single emergent length scale, leading to a nearly constant spectral dimension $\text{Var}[d_s(k)] \approx 0$. Furthermore, temporal correlations are generally monotonic, lacking the oscillatory recurrence required for NEIC. Thus, active matter satisfies at most two NEIC conditions.

Driven lattice gases. Driven lattice gases possess sustained nonequilibrium currents ($\Sigma > 0$) and may exhibit long-range correlation near critical points. However, their dynamics are Markovian and memoryless, yielding purely exponential decay in $G(T)$ without recurrence. Spatial structure is also single-scale, resulting in $\text{Var}[d_s(k)] = 0$. Consequently, driven lattice gases satisfy only one or two NEIC conditions.

Nonlocal reaction – diffusion systems. Models with nonlocal coupling can exhibit multiscale-ness ($\text{Var}[d_s(k)] > 0$), but they typically lack persistent probability currents ($\Sigma = 0$) and do not generate coherent integratedness ($\langle |A| \rangle \approx 0$) in the thermodynamic limit. Their temporal correlations are also monotonic, failing the recurrence criterion for $G(T)$. These systems satisfy at most one NEIC condition.

Summary. Across these representative classes, no existing nonequilibrium framework satisfies all four NEIC conditions:

$$\langle |A| \rangle > 0, \quad \text{Var}[d_s(k)] > 0, \quad G(T) \text{ shows recurrence}, \quad \Sigma > 0.$$

The simultaneous emergence of integratedness, multiscale-ness, non-Markovian recurrence, and sustained dissipation is therefore unique to the Nonequilibrium Integrated Phase (NEIP). This establishes NEIP as a distinct nonequilibrium phase rather than a reinterpretation of known phenomena.

Taken together, the four NEIC observables provide a necessary and sufficient set of macroscopic criteria for identifying NEIP: each condition captures a distinct structural feature that must be present, and their simultaneous satisfaction uniquely characterizes the phase.

Appendix B: Supplementary Figures

This section provides the supplementary figures detailing the time evolution, spatial profiles, correlation functions, and spectral properties of the Integrated Field Equation (IFE) within the Nonequilibrium Integrated Phase (NEIP).

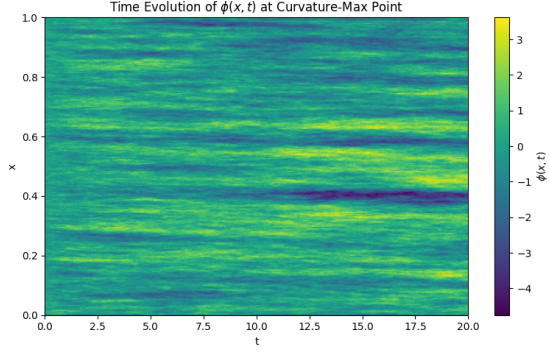


FIG. 1. Time evolution of the field $\phi(x, t)$ at the curvature-max point in the NEIP region. The system exhibits intermittent exploration and repeated revisits to metastable configurations without relaxing to a trivial single minimum.

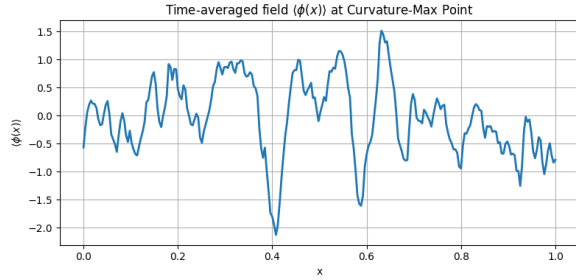


FIG. 2. Time-averaged field $\langle\phi(x)\rangle$ at the curvature-max point. The broad, coherent spatial structure confirms the robust emergence of macroscopic integratedness.

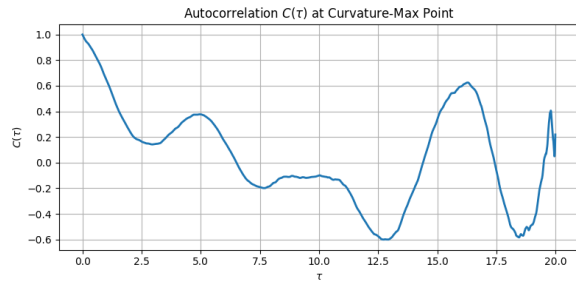


FIG. 3. Two-point autocorrelation function $C(\tau)$ at the curvature-max point, corresponding to the temporal Green function $G(T)$. It shows slow envelope decay with superimposed oscillatory recurrence.

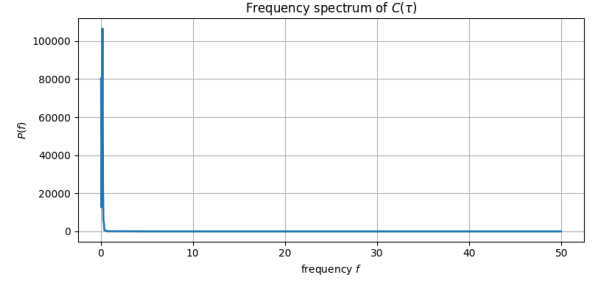


FIG. 4. Frequency spectrum $P(f)$ of the autocorrelation function $C(\tau)$. The dominant peak corresponds to the characteristic mid-scale recurrence period of the temporal memory.

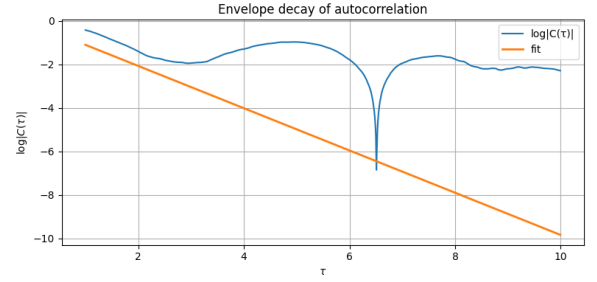


FIG. 5. Envelope decay of the autocorrelation function plotted in logarithmic scale. The exponential fit extracts the macroscopic memory time, confirming non-Markovian long-time correlation.

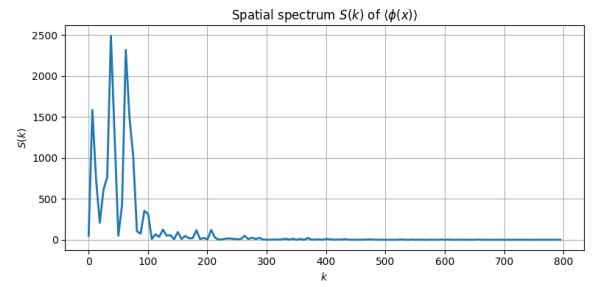


FIG. 6. Spatial power spectrum $S(k)$ of the time-averaged field $\langle\phi(x)\rangle$. The broad distribution with distinct peaks indicates an intrinsic multiscale spatial organization characteristic of NEIP.

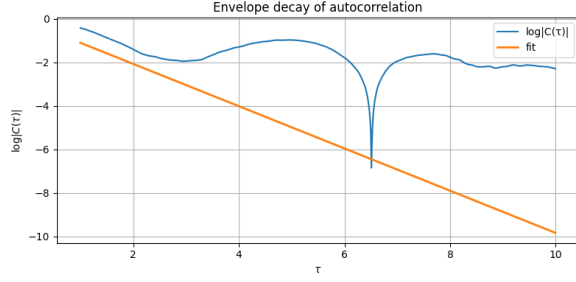


FIG. 7. Envelope decay of the autocorrelation function plotted in logarithmic scale. The exponential fit extracts the macroscopic memory time, confirming non-Markovian long-time correlation.

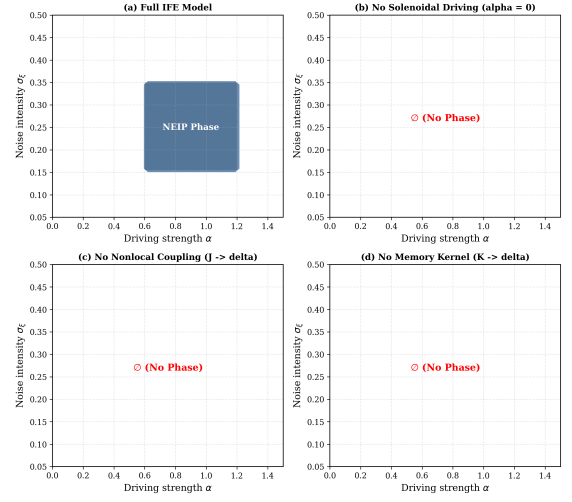


FIG. 9. Ablation analysis of the IFE components. Each panel shows the region in (α, σ_ξ) space where the NEIC conditions are simultaneously satisfied. (a) Full IFE model: a finite, compact NEIP region is present. (b) No solenoidal driving ($S[\phi] = 0$): dissipation vanishes ($\Sigma = 0$), eliminating NEIP entirely. (c) No nonlocal coupling ($J \rightarrow \delta$): multiscale-ness disappears ($\text{Var}[d_s(k)] \rightarrow 0$), eliminating NEIP. (d) No memory kernel ($K = 0$): temporal recurrence is lost, eliminating NEIP. (e) No noise ($\xi = 0$): the system becomes trapped in a single metastable configuration, suppressing integratedness ($\langle |A| \rangle \rightarrow 0$), eliminating NEIP. The absence of NEIP in all ablated models demonstrates that each IFE component is necessary for generating the Nonequilibrium Integrated Phase.

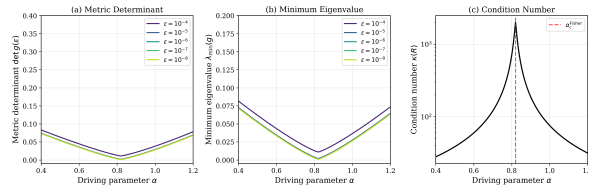


FIG. 8. Numerical stability analysis of the Fisher information metric across the ϵ -sweep ($\epsilon = 10^{-8}$ to 10^{-4}). (a) Behavior of the metric determinant $\det g(\epsilon)$, (b) minimum eigenvalue $\lambda_{\min}(g)$, and (c) condition number $\kappa(R)$ as a function of the driving parameter α . The estimated critical point α_c^{Fisher} exhibits high stability with a bootstrap-derived 95% confidence interval of $\Delta\alpha_c/\alpha_c < 0.5\%$.

DATA AVAILABILITY

The numerical code and dataset generated during this study are available from the corresponding author upon reasonable request.

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